Experiment-1

Aim: Study of Geiger-Muller Counter and counting statistics by γ-ray source

Abstract

The characteristic curve of the Geiger-Muller region and the counting statistics of a given source, i.e. Barium (133), have been studied using the Geiger-Muller counter.

Theory

Due to the fact that Nuclear reactions deal with length scales of the order of 10^{-15}m detection of these reactions, which include the products and in times trajectories of these product, have to have a way of amplifying any interaction the product has with matter.

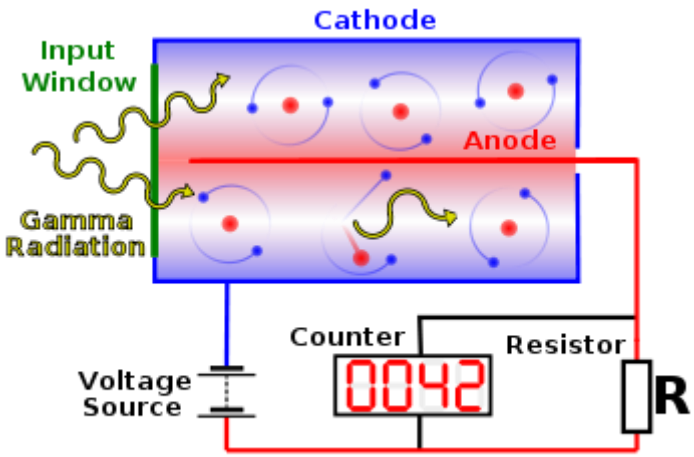
There are different types of detectors used depending upon the requirement of the experiment. They are classified as

1) The gas filled detectors which include
    - Ionization chamber
    - Proportional chamber
    - Geiger-Muller counter
2) Semiconductor detector
3) Scintillation counters
4) Cerenkov counters
5) Photographic emulsions
6) Cloud chambers
7) Bubble chambers
8) Spark chambers

In this experiment we will be using the Geiger Muller counter

As the gamma rays are themselves not ionizing particles their detection is done using secondary ionizations which they produce in suitable interactions.

Geiger-Muller counter



A Geiger–Müller tube consists of a tube filled with a low-pressure (~0.1 Atm) inert gas such as helium, neon or argon (usually neon), in some cases in a Penning mixture, and an organic vapor or a halogen gas. The tube contains electrodes, between which there is a potential difference of several hundred volts, but no current flowing. The walls of the tube are either entirely metal or have their inside surface coated with a conductor to form the cathode while the anode is a wire passing up the center of the tube.

When ionizing radiation passes through the tube, some of the gas molecules are ionized, creating positively charged ions, and electrons. The strong electric field created by the tube's electrodes accelerates the ions towards the cathode and the electrons towards the anode. The ion pairs gain sufficient energy to ionize further gas molecules through collisions on the way, creating an avalanche of charged particles.

This results in a short, intense pulse of current which passes (or cascades) from the negative electrode to the positive electrode and is measured or counted.

The number of pulses per second measures the intensity of the radiation field. Some Geiger counters display an exposure rate, but this does not relate easily to a dose rate as the instrument does not discriminate between radiations of different energies.

The usual form of GM tube is an end-window tube. This type is so-named because the tube has a window at one end through which ionizing radiation can easily penetrate. The other end normally has the electrical connectors. There are two types of end-window tubes: the glass-mantle type and the mica window type. The glass window type will not detect alpha radiation

since it is unable to penetrate the glass, but is usually cheaper and will usually detect beta radiation and X-rays. The mica window type will detect alpha radiation but is more fragile.

Most tubes will detect gamma radiation, and usually beta radiation above about 2.5 MeV. Geiger–Müller tubes will not normally detect neutrons since these do not ionise the gas. However, neutron-sensitive tubes can be produced which either have the inside of the tube coated with boron or contain boron trifluoride or helium-3 gas. The neutrons interact with the boron nuclei, producing alpha particles or with the helium-3 nuclei producing hydrogen and tritium ions and electrons. These charged particles then trigger the normal avalanche process.

Although most tubes will detect gamma radiation, standard tubes are relatively inefficient, as most gamma photons will pass through the low density gas without interacting. Using the heavier noble gases krypton or xenon for the fill effects a small improvement, but dedicated gamma detectors use dense cathodes of lead or stainless steel in windowless tubes. The dense cathode then interacts with the gamma flux, producing high-energy electrons, which are then detected.

The ideal GM tube must produce a single pulse on entry of a single particle. It must not give any spurious pulses, and must recover quickly to the passive state. Unfortunately for these requirements, the positive argon ions that eventually strike the cathode become neutral argon atoms in an excited state by gaining electrons from the cathode. The excited atoms return to the ground state by emitting photons and these photons cause avalanches and hence spurious pulse discharge. Quenching of this process is thus important because a single particle entering the tube is counted by a single discharge, and so the tube is unable to re-set and detect another particle until the discharge has been stopped. Also, the tube is damaged by prolonged discharges.

External quenching uses external electronics to remove the high voltage between the electrodes. Self-quenching or internal-quenching tubes stop the discharge without external assistance, by the addition of a small amount of a polyatomic organic vapor such as butane or ethanol; or alternatively a halogen such as bromine or chlorine.

If a poor diatomic gas quencher is introduced to the tube, the positive argon ions, during their motion toward the cathode, would have multiple collisions with the quencher gas molecules and transfer their charge and some energy to them. Thus, neutral argon atoms would be produced and the quencher gas ions in their turn would reach the cathode, gain electrons there from, and move into excited states which would decay by photon emission, producing tube discharge. However, effective quencher molecules, when excited, lose their energy not by photon emission, but by dissociation into neutral quencher atoms. No spurious pulses are thus produced.

Gaussian distribution

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

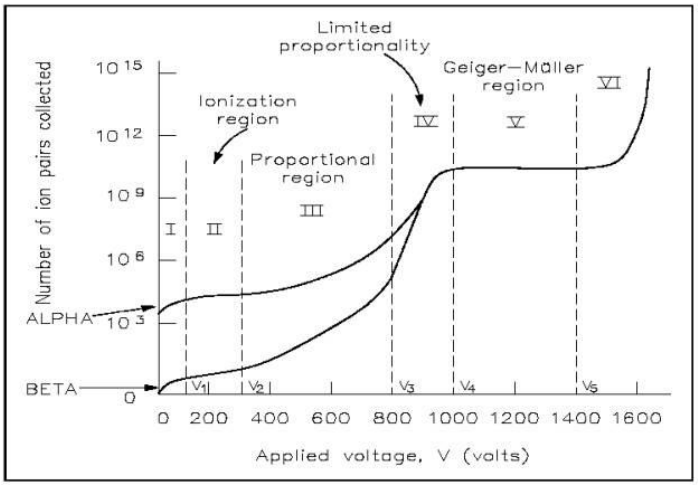
The normal distribution is considered the most prominent probability distribution in statistics. There are several reasons for this: First, the normal distribution is very tractable analytically, that is a large number of results involving this distribution can be derived in explicit form. Second, the normal distribution arises as the outcome of the central limit theorem, which states that under mild conditions the sum of a large number of random variables is distributed approximately normally. Finally, the "bell" shape of the normal distribution make it a convenient choice for modeling a large variety of random variables encountered in practice.

For this reason, the normal distribution is commonly encountered in practice, and is used throughout statistics, natural sciences, and social sciences as a simple model for complex phenomena. For example, the observational error in an experiment is usually assumed to follow a normal distribution, and the propagation of uncertainty is computed using this assumption. Note that a normally-distributed variable has a symmetric distribution about its mean. Quantities that grow exponentially, such as prices, incomes or populations, are often skewed to the right, and hence may be better described by other distributions, such as the log-normal distribution or Pareto distribution. In addition, the probability of seeing a normally-distributed value that is far (i.e. more than a few standard deviations) from the mean drops off extremely rapidly. As a result, statistical inference using a normal distribution is not robust to the presence of outliers (data that is unexpectedly far from the mean, due to exceptional circumstances, observational error, etc.). When outliers are expected, data may be better described using a heavy-tailed distribution such as the Student's t-distribution.

From a technical perspective, alternative characterizations are possible, for example:

- The normal distribution is the only absolutely continuous distribution all of whose cumulants beyond the first two (i.e. other than the mean and variance) are zero.
- For a given mean and variance, the corresponding normal distribution is the continuous distribution with the maximum entropy.

Procedure



The characteristic curve was plotted by placing the source Barium (133) at slot 4 in the apparatus and the count was measured twice while changing the high voltage supplied to the dynode and this graph was plotted.

Now the source was placed at the bottom most slot and the count was measured for 200 turns keeping the voltage in the Geiger region (650V) for 5s. This count v/s the frequency of the counts was plotted and the process repeated for different slots (third and first) and the results compared.

Results

Table 1: Study of characteristic curve for Geiger Muller region.

DC Voltage (Volt)	Number of counts				Average number of counts
	Run no. 1	Run no. 2	Run no. 3	Run no. 4	
350	56	61	47	43	51.75
400	61	56	52	56	56.25
450	49	70	45	55	54.75
500	57	49	57	60	55.75
550	49	60	63	71	60.75
600	51	54	53	60	54.5

650	50	61	56	53	55
700	56	66	58	49	57.25
750	65	48	48	56	54.25
800	68	57	66	50	60.25
850	49	68	55	56	57
900	56	55	54	46	52.75
950	50	61	45	63	54.75
1000	72	48	57	60	59.25
1050	64	66	78	67	68.75
1100	137	123	180	205	161.25

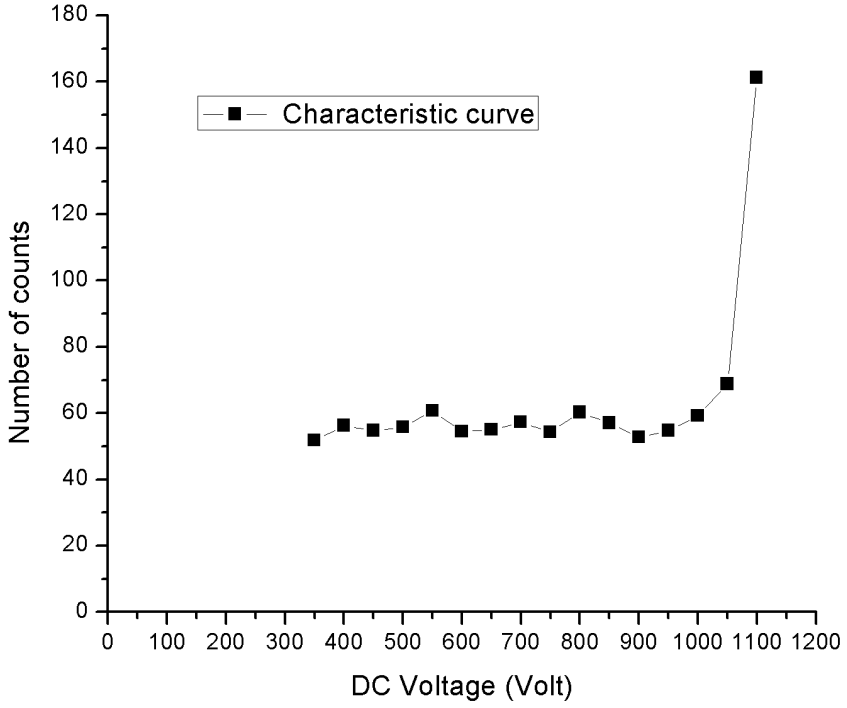


Table 2: Study of distribution curve of the counts for source to be at the maximum distance possible in the stand from Geiger tube.

Number of counts	Frequency of counts
0	1
1	5
2	22
3	31
4	49
5	34
6	28
7	16
8	8
9	3
10	1
11	2

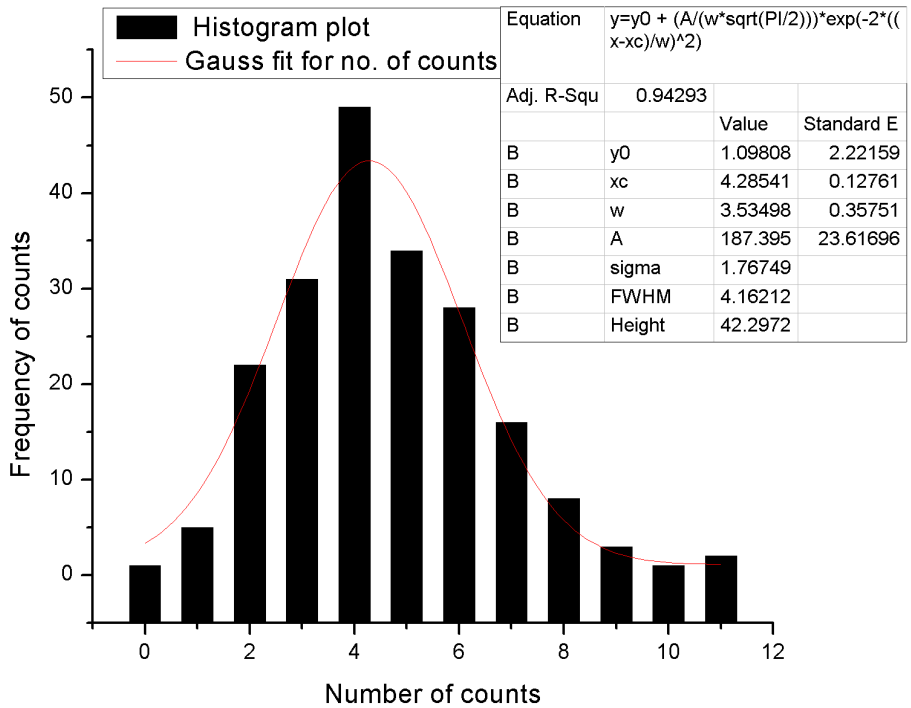
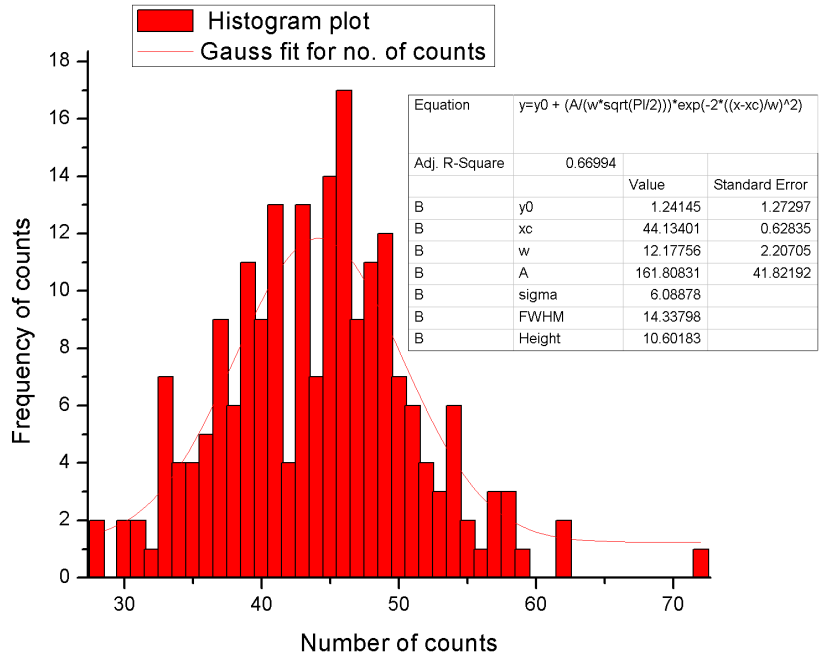


Table 3: Study of distribution curve of the counts for source to be at the 3rd slot in the stand from Geiger tube.

Number of counts	Frequency of counts
28	2
30	2
31	2
32	1
33	7
34	4
35	4
36	5

37	9
38	6
39	11
40	9
41	13
42	4
43	13
44	7
45	14
46	17
47	9
48	11
49	12
50	7
51	6
52	4
53	3
54	6
55	2
56	1
57	3
58	3
59	1
62	2

72	1



Conclusions

The characteristic curve got from the experiment matched the theory and the Geiger region was found to be from 350-1000V.

The count v/s frequency of count curve for the source being placed in the lowermost slot approximated to a Gaussian much better than the one where the source was placed in the third slot, though even this had the looks of a Gaussian and would have become one had more counts been taken. The third curve for the source in the first slot was not plotted as the number of counts taken was very few for a favorable statistics.

Therefore we can conclude from this that the energies of the gamma radiation being emitted forms a normal distribution.

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